



4866 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam pre-imaging				
	1	NIRCam mosaic	NIRCam Imaging	(1) IC348
NIRSpec MOS placeholder				
	2	ic348a2	NIRSpec MultiObject Spectroscopy	(4) mpt2
	3	ic348b2	NIRSpec MultiObject Spectroscopy	(4) mpt2
	4	ic348c2	NIRSpec MultiObject Spectroscopy	(4) mpt2
	5	ic348d4	NIRSpec MultiObject Spectroscopy	(5) mpt3

ABSTRACT

A Cycle 1 program obtained NIRCam images for a small field (6'x4') in the center of IC 348, which is a nearby star-forming cluster (300 pc, 5 Myr). Those data were used to identify candidates for new free-floating planetary-mass objects (PMOs), which were then observed spectroscopically with NIRSpec (1-5 μm) to confirm their cool nature. That program discovered three new PMOs, the faintest of which may be the least massive object that has been directly imaged and spectroscopically confirmed to date ($\sim 3\text{-}4$ Jupiter masses). In addition, two of the three PMOs exhibit several strong absorption features from an unidentified aliphatic hydrocarbon, the presence of which is unexplained by models of brown dwarfs and giant planets. The only atmospheres in which similar hydrocarbons have been previously detected are those of Saturn and Titan.

We propose a wider (16'x20') and deeper survey for PMOs in IC 348 that reaches the mass of Jupiter. The objectives are as follows: 1) improve the statistical constraints on the minimum mass of brown dwarfs and the frequency of ejected giant planets, 2) compare the mass functions of PMOs in IC 348 and the Orion Nebula Cluster to test whether the frequency of PMOs varies with stellar density, 3) assess whether the frequency of PMOs in IC 348 varies with distance from the center of the cluster, as might be expected if some of them are ejected planets, and 4) test scenarios for the origin of the unidentified hydrocarbon by determining whether they are found in all PMOs below a certain mass or temperature, or whether their presence is correlated with other factors (e.g., UV flux from the most massive star in the cluster).

OBSERVING DESCRIPTION

NIRCam images will be obtained in a mosaic with 9 rows and 3 columns in the IC348 star-forming cluster, corresponding to a 16'x20' field. The images will be taken with the FULLBOX dither pattern, four dithers, and four filters (F162M, F182M, F360M, F444W). The readout pattern will be SHALLOW4 with 5 groups, resulting in a total exposure time of 1031 sec among the four dithers at a given position and filter. The NIRCam data will serve as pre-imaging for NIRSpec MOS observations. The APT file includes four placeholder MOS configurations for NIRSpec, each of which uses the PRISM disperser, NRSIR2RAPID readout pattern, 100 groups, and three dithers, corresponding to a total exposure time of 4420 sec. The NIRSpec observations include an implicit requirement to occur after the NIRCam imaging by 60 days or more.

Proposal 4866 - Targets - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	IC348	RA: 03 44 31.0000 (56.1291667d) Dec: +32 08 30.00 (32.14167d) Equinox: J2000 Comments: Category=Stellar Cluster Description=[Young star clusters] Extended=NO		
(2)	mpt	RA: 03 44 40.8880 (56.1703667d) Dec: +32 12 57.79 (32.21605d) Equinox: J2000 Comments: Description=[]		
(3)	mpt5	RA: 03 44 41.7892 (56.1741217d) Dec: +32 12 46.86 (32.21302d) Equinox: J2000 Comments: Description=[]		
(4)	mpt2	RA: 03 44 40.6588 (56.1694117d) Dec: +32 12 54.67 (32.21519d) Equinox: J2000 Comments: Description=[]		
(5)	mpt3	RA: 03 44 40.6856 (56.1695233d) Dec: +32 12 54.43 (32.21512d) Equinox: J2000 Comments: Description=[]		

Proposal 4866 - Observation 1 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Observation	Proposal 4866, Observation 1: NIRCam mosaic				Thu Jan 16 19:00:38 GMT 2025
	Diagnostic Status: Warning				
Observing Template: NIRCam Imaging					
Diagnostics	(NIRCam mosaic (Obs 1)) Warning (Form): This observation is split across multiple visits using multiple filters. Not selecting the sequence option may result in execution of the visits in a non-numerical order and is not recommended.				
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:13) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:14) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:15) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:16) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:17) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:18) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:19) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:20) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:21) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:22) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:23) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:24) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:25) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:26) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 1:27) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	IC348	RA: 03 44 31.0000 (56.1291667d) Dec: +32 08 30.00 (32.14167d) Equinox: J2000 Comments: Category=Stellar Cluster Description=[Young star clusters] Extended=NO		

Proposal 4866 - Observation 1 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Template	Module		Subarray				Target Placement			
	ALL		FULL				Module Gap			
Mosaic	Rows		Columns		Row Overlap %		Column Overlap %		Row shift (deg)	
	9		3		0.0		0.0		0.0	
Dithers			Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size	
	1		FULLBOX		4		STANDARD		1	
Spectral Elements	#		Short Filter		Long Filter		Readout Pattern		Groups/Int	
	1		F162M+F150W2		F360M		SHALLOW4		5	
	2		F182M		F444W		SHALLOW4		5	
Special Requirements										

Proposal 4866 - Observation 2 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Observation	Proposal 4866, Observation 2: ic348a2										Thu Jan 16 19:00:38 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	mpt2	RA: 03 44 40.6588 (56.1694117d) Dec: +32 12 54.67 (32.21519d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map		Spectral Overlap Threshold		
	MSATA	false	No	MSA Center	primary2 (26 sources)		jwst-nirspec-prism		1.5		
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	6188	56.277819	32.266547	21.275	1	6860	56.231202	32.277117	21.602	
	1	6203	56.277658	32.264360	21.565	1	6967	56.243002	32.229122	21.735	
	1	6299	56.274890	32.258160	21.604	1	7182	56.219161	32.251230	21.626	
	1	6787	56.251757	32.238049	21.421	1	7214	56.222311	32.237809	21.327	
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	56.251193333333 33 Degrees 32.249469444444 44 Degrees	210.98298643929 12			3	6	4445.023

Proposal 4866 - Observation 2 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Special Requirements	MSA Scheduled Aperture PA 210.9394 to 210.9394 Degrees (V3 72.364845 to 72.364845) 2 After 1 by 60.0 Days to <None specified>
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Proposal 4866 - Observation 3 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Observation	Proposal 4866, Observation 3: ic348b2											Thu Jan 16 19:00:38 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(4)	mpt2	RA: 03 44 40.6588 (56.1694117d) Dec: +32 12 54.67 (32.21519d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold				
	MSATA	false	No	MSA Center	primary2 (26 sources)		jwst-nirspec-prism	1.5				
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	2381	56.071087	32.221314	21.044	1	2865	56.040468	32.225034	21.511		
	1	2459	56.053495	32.252761	22.455	1	2944	56.028523	32.242053	22.709		
	1	2636	56.055509	32.223173	22.611	1	3401	56.084377	32.215459	22.568		
	1	2714	56.047358	32.231397	22.074	1	5291	56.085038	32.248023	21.774		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time	
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	56.062181666666 67 Degrees 32.239744444444 45 Degrees	212.88135018799 278			3	9	4445.023	

Proposal 4866 - Observation 3 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Special Requirements	MSA Scheduled Aperture PA 212.9386 to 212.9386 Degrees (V3 74.364 to 74.364) 3 After 1 by 60.0 Days to <None specified>
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Proposal 4866 - Observation 4 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Observation	Proposal 4866, Observation 4: ic348c2 Diagnostic Status: Warning Observing Template: NIRSpec MultiObject Spectroscopy											Thu Jan 16 19:00:38 GMT 2025
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous				
	(4)	mpt2	RA: 03 44 40.6588 (56.1694117d) Dec: +32 12 54.67 (32.21519d) Equinox: J2000									Comments: Description=[]
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method		HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map		Spectral Overlap Threshold		
	MSATA		false	No	MSA Center	primary2 (26 sources)		jwst-nirspec-prism		1.5		
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	655	56.022777	32.073571	22.815	1	935	56.023559	32.018385	22.640		
	1	727	56.018143	32.070110	22.524	1	1299	55.978162	32.038771	21.272		
	1	827	56.009789	32.071398	22.537	1	1349	55.975035	32.035279	22.129		
	1	883	56.002303	32.079227	22.322	1	1517	56.016279	32.033632	22.147		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time	
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	56.010934583333 34 Degrees 32.047344444444 44 Degrees	216.85909916790 05			3	3	4420.434	

Proposal 4866 - Observation 4 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Special Requirements	MSA Scheduled Aperture PA 216.9433 to 216.9433 Degrees (V3 78.36877 to 78.36877) 4 After 1 by 60.0 Days to <None specified>
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Proposal 4866 - Observation 5 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Observation	Proposal 4866, Observation 5: ic348d4											Thu Jan 16 19:00:38 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(5)	mpt3	RA: 03 44 40.6856 (56.1695233d) Dec: +32 12 54.43 (32.21512d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images		Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
	MSATA	false	No		MSA Center	primary3 (4 sources)		jwst-nirspec-prism		1.5		
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	140	56.033653	32.134906	21.809	1	892	55.989526	32.108972	21.313		
	1	248	56.020353	32.148826	22.218	1	914	55.979828	32.127475	22.265		
	1	570	55.988854	32.172358	21.954	1	1051	55.971441	32.120826	21.688		
	1	615	56.008447	32.116300	22.489	1	1090	55.965901	32.123507	21.594		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time	
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	55.998027916666 665 Degrees 32.134958333333 33 Degrees	216.90871149443 56			3	3	4420.434	

Proposal 4866 - Observation 5 - A Survey for Free-floating Brown Dwarfs and Planets Down to the Mass of Jupiter

Special Requirements	Aperture PA Range 217.000 to 217.000 Degrees (V3 78.4254303 to 78.4254303) [MSA Selected] MSA Scheduled Aperture PA 217.0000 to 217.0000 Degrees (V3 78.4254303 to 78.4254303) 5 After 1 by 60.0 Days to <None specified>
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